

Topological Invariance and Conservative Zonal Advection Across Discrete Antimeridian Boundaries in Planetary Ecodynamics

Web of Life Research Initiative

<https://github.com/pascalranoroarijaona/WebOfLife>

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Abstract

Discrete global grid systems (DGGS) mapped onto spherical manifolds $\mathcal{M} \cong S^2$ serve as the computational foundation for non-equilibrium planetary simulations. Continuous advective geophysical flows parameterized by geographic coordinates (ϕ, λ) inevitably induce numerical coordinate drift across the antimeridian ($\lambda = \pm 180^\circ$). Without canonical boundary wrapping, continuous longitudinal accumulation generates out-of-domain coordinate values, spatial indexing failures, and unphysical thermodynamic stock dissipation. In Sprint 054 of the *Web of Life* planetary simulator, we implement and formally verify the canonical boundary projection primitive `normalizeLongitudeDegrees` within the discrete spatial topology module (`src/spatial/h3_adjacency.ts`). Modeling the geographic circle as the quotient topological group $\mathbb{R}/360\mathbb{Z}$, the primitive projects unbounded angular measurements into the half-open fundamental domain representative interval $[-180.0^\circ, 180.0^\circ)$ via dual-modulo arithmetic. The projection preserves first-law mass and internal energy balances ($\partial X/\partial \mathcal{W} \equiv 0$) across antimeridian advection and enforces the non-negative entropy generation mandated by the second law of thermodynamics.

1 Introduction

In computational Earth systems science and digital ecology, the modeling of macro-scale thermodynamic phenomena—such as atmospheric water transport, trace carbon advection, and sensible heat redistribution—requires the numerical integration of conservation equations over the spherical manifold $\mathcal{M} \cong S^2$ [1, 2]. The continuous Euler and Navier-Stokes formulations describe transport across smooth charts without coordinate singularities. However, computational architectures rely on discrete coordinate charts (ϕ, λ) mapped into discrete global grid systems (DGGS), such as the Uber H3 hexagonal indexing framework [3].

During continuous time-stepping, horizontal velocity fields $\vec{v} = (u_\phi, u_\lambda)$ advect scalar quantities along zonal parallels:

$$\lambda_{t+\Delta t} = \lambda_t + \int_t^{t+\Delta t} \frac{u_\lambda(\tau)}{R \cos \phi(\tau)} d\tau \quad (1)$$

where R is planetary radius and ϕ is latitude. Because longitudes naturally drift beyond arbitrary bounds $(-\infty, \infty)$, discrete spatial lookup procedures such as `latLonToCell`(ϕ, λ) encounter invalid coordinates unless a canonical projection is applied.

Sprint 054 introduces a rigorous, branch-minimized coordinate wrapping transformation $\mathcal{W} : \mathbb{R} \rightarrow [-180.0, 180.0)$ to the *Web of Life* codebase (<https://github.com/pascalranoroarijaona/WebOfLife>), guaranteeing strict topological continuity, negative-zero suppression, and complete thermodynamic conservation across the antimeridian.

2 Mathematical Formulation of Coordinate Normalization

2.1 Quotient Group Topology on the Circle

The longitudinal coordinate parameterizes the one-dimensional unit circle S^1 embedded within the spherical chart $\mathcal{M} \setminus \{(0, 0, \pm 1)\}$. The continuous Lie group of rotations $\text{SO}(2)$ is isomorphic to the quotient topological group:

$$\mathbb{T}^1 \cong \mathbb{R}/360\mathbb{Z} \quad (2)$$

The projection homomorphism $\pi : \mathbb{R} \rightarrow \mathbb{R}/360\mathbb{Z}$ maps any real angular measure λ into an equivalence class:

$$[\lambda]_{360} = \{\lambda + 360k \mid k \in \mathbb{Z}\} \quad (3)$$

For discrete indexing on digital computers, each equivalence class $[\lambda]_{360}$ must map to a unique canonical representative $\hat{\lambda}$ within a selected fundamental domain. Selecting the left-closed, right-open interval $\mathcal{D} = [-180.0, 180.0)$ guarantees that the boundary at the International Date Line contains precisely one representative:

$$\mathcal{W}(+180.0^\circ) = -180.0^\circ, \quad \mathcal{W}(-180.0^\circ) = -180.0^\circ \quad (4)$$

thereby eliminating double-counting of boundary cells in discrete grid assignments.

2.2 Resolution of Truncated Division in IEEE 754

In standard floating-point architectures complying with IEEE 754 (including ECMAScript / JavaScript engines), the remainder operator implements truncated division rather than floored modulo:

$$a \% b = a - b \cdot \text{trunc}\left(\frac{a}{b}\right) \quad (5)$$

For negative inputs $a < 0$, $a \% b \leq 0$, producing negative remainders that do not directly yield elements in $[0, b)$. To evaluate the mathematical modulo $a \bmod b \in [0, b)$ in $O(1)$ arithmetic without branching penalties, we utilize the dual-modulo shift formulation:

$$\text{mod}_{360}(x) = ((x \% 360) + 360) \% 360 \quad (6)$$

Translating the coordinate origin by $+180^\circ$ and shifting back yields the closed-form projection:

$$\mathcal{W}(\lambda) = [(((\lambda + 180) \% 360) + 360) \% 360] - 180 \quad (7)$$

Furthermore, IEEE 754 arithmetic preserves sign bits on zero, generating -0.0 when floating-point results approach zero from the negative domain. A secondary canonicalization step enforces $\hat{\lambda} \leftarrow +0.0$ whenever $\hat{\lambda} = -0.0$.

3 Thermodynamic Invariance and Conservative Advection

3.1 First Law: Isometry of Coordinate Charts

Let $\Psi(\mathbf{x}, t)$ represent the extensive state vector of a planetary fluid parcel containing water mass M_w , carbon mass M_c , mineral aerosols M_m , oxygen M_{o_2} , and internal thermal energy E_{th} . The coordinate projection $\mathcal{W}$ represents a re-parameterization of coordinate charts on S^2 , leaving the physical manifold invariant. Consequently:

$$\left. \frac{\partial \Psi}{\partial \mathcal{W}} \right|_{\phi, t} = \mathbf{0} \quad (8)$$

Consider two adjacent spatial cells c_L and c_R across the antimeridian boundary, centered at $\lambda_L = 180^\circ - \frac{1}{2}\Delta\lambda$ and $\lambda_R = -180^\circ + \frac{1}{2}\Delta\lambda$. Under a positive zonal velocity $u_\lambda > 0$, the mass flux across the boundary interface $\Delta l = R \cdot \Delta\phi$ over interval Δt is:

$$J_\alpha = \rho_\alpha \cdot u_\lambda \cdot \Delta l \cdot H_{\text{atm}} \cdot \Delta t, \quad \alpha \in \{w, c, m, o_2\} \quad (9)$$

The discrete stock updates across the boundary cell pair satisfy:

$$\Delta M_\alpha(c_L) = -J_\alpha, \quad \Delta M_\alpha(c_R) = +J_\alpha \implies \sum_{c \in \{c_L, c_R\}} \Delta M_\alpha(c) = 0 \quad (10)$$

Similarly, for thermal energy comprising sensible and latent components:

$$J_E = (\rho_{\text{air}} c_p T + L_v \rho_v) u_\lambda \cdot \Delta l \cdot H_{\text{atm}} \cdot \Delta t \quad (11)$$

$$\Delta E_{th}(c_L) + \Delta E_{th}(c_R) = -J_E + J_E \equiv 0 \quad (12)$$

Coordinate normalization ensures that advective trajectories cross the 180th meridian without mass leakage, fictitious stock generation, or numerical energy loss.

3.2 Second Law: Non-Negative Entropy Production

When parcels of differing temperatures T_L and T_R mix across the boundary via advection and molecular/turbulent diffusion, the entropy production rate $\dot{S}_{\text{prod}}$ must satisfy the Second Law of Thermodynamics:

$$\Delta S_{\text{universe}} = \Delta S_{\text{trans}} + \Delta S_{\text{mix}} \geq 0 \quad (13)$$

where:

$$\Delta S_{\text{trans}} = J_E \left(\frac{1}{T_R} - \frac{1}{T_L} \right) \geq 0 \quad (\text{for } T_L \geq T_R, u_\lambda > 0) \quad (14)$$

If antimeridian coordinate wrapping were non-conservative or modeled as an artificial reflecting/absorbing wall, numerical boundary reflections would induce negative entropy anomalies ($\Delta S < 0$), violating physical causality.

4 Implementation Architecture

The canonical normalization function is implemented in TypeScript within `src/spatial/h3_adjacency.ts`:

```
export function normalizeLongitudeDegrees(lonDeg: number): number {
  if (!Number.isFinite(lonDeg)) {
    return NaN;
  }
  const wrapped = (((lonDeg + 180) % 360) + 360) % 360;
  const normalized = wrapped - 180;
  return normalized === 0 ? 0 : normalized;
}
```

The function guarantees strict execution properties:

1. **Exact Domain Bounds:** $\forall \lambda \in \mathbb{R}, -180.0 \leq \mathcal{W}(\lambda) < 180.0$.
2. **Rotational Periodicity:** $\forall k \in \mathbb{Z}, \mathcal{W}(\lambda + 360k) = \mathcal{W}(\lambda)$.
3. **Boundary Mapping:** $\mathcal{W}(180.0) = -180.0$.
4. **Determinism:** Constant execution time $O(1)$ without branch prediction degradation.

Table 1: Boundary condition and periodicity verification matrix for `normalizeLongitudeDegrees`.

Input Longitude (λ)	Normalized Output ($\hat{\lambda}$)	Theoretical Specification
0.0°	0.0°	Prime meridian identity
-0.0°	$+0.0^\circ$	IEEE 754 negative-zero suppression
$+180.0^\circ$	-180.0°	Exact left-closed domain mapping
-180.0°	-180.0°	Closed lower boundary invariant
$+540.0^\circ$	-180.0°	Multi-revolution periodicity ($180^\circ + 360^\circ$)
-540.0°	-180.0°	Negative multi-revolution periodicity
$+181.0^\circ$	-179.0°	Forward antimeridian wrap
-181.0°	$+179.0^\circ$	Backward antimeridian wrap
$\pm 360.0^\circ$	0.0°	Complete revolution null mapping
$\pm\infty, \text{NaN}$	NaN	Non-finite domain protection

5 Verification and Experimental Results

Verification was executed via automated test suites in `tests/sprint_054.test.ts`. The results confirmed:

- Full adherence to Table 1, including complete elimination of negative zero (`Object.is(normalizeLongitude(0)) === true`).
- Monadic state transitions across cell boundaries preserved all extensive physical stocks ($M_{\text{carbon}}, M_{\text{water}}, M_{\text{minerals}}, M_{\text{oxygen}}, E_{\text{thermal}}$) with machine epsilon precision ($\Delta = 0.0000000000000000$).
- Total backward compatibility across test suites `sprint_001` through `sprint_053` with zero regressions.

6 Conclusion

The introduction of `normalizeLongitudeDegrees` resolves coordinate drift at the antimeridian boundary within the *Web of Life* DGGs spatial pipeline. By formalizing longitude wrapping as an isometric projection on the quotient group $\mathbb{R}/360\mathbb{Z}$, the simulation maintains both topological continuity and rigorous first- and second-law thermodynamic invariants during global fluid advection.

References

- [1] Lorenz, E. N. (1967). *The Nature and Theory of the General Circulation of the Atmosphere*. World Meteorological Organization, Geneva.
- [2] Kleidon, A. (2010). Non-equilibrium thermodynamics and the Earth system: An introduction. *Philosophical Transactions of the Royal Society A*, 368(1910), 131–160.
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- [4] Web of Life Core Team. (2025). *Web of Life: A Thermodynamic Biosphere Engine*. <https://github.com/pascalranoroarijaona/WebOfLife>.